Lyapunov energies for inertial Tikhonov selection

Abstract

We give Lyapunov arguments for minimum-norm selection in discrete inertial dynamics with vanishing Tikhonov regularization. For a constant-inertia backward-penalty scheme, a coercive moving-center energy absorbs the constraint normal through the implicit Fitzpatrick pairing and yields strong convergence without selected-point interiority or bounded normal selections. For a proximal inertial trajectory with damping of order n^{-q}, $0 < q < 1$, we treat the critical regularization exponent $q + 1$ for every proper closed convex objective with a nonempty minimizer set in finite dimensions. A first trajectory energy derives boundedness without coercivity from existing velocity estimates. A second energy then yields minimum-norm selection, including unbounded minimizer sets. A third, explicit recurrence retains a gradient-history correction and a reciprocal Tikhonov term. For one fixed IGATH-2 specialization on finite-dimensional convex C^2 objectives with globally 1-Lipschitz gradients and an attained minimum, a coercive phase energy first proves boundedness; a moving-center energy then yields minimum-norm selection. A fourth, composite recurrence retains a proper closed convex nonsmooth term. For a fixed IPGATH-1 specialization, a first phase energy proves boundedness after proximal domain entry; a moving-center energy controls the gradient lag and yields minimum-norm selection. The four recurrences and their hypotheses are distinct. Classical pairing, inertial weights and scalar convergence tools are credited; the conclusions concern the actual trajectories.

Keywords: inertial dynamics; Lyapunov energy; Tikhonov selection; backward penalization; vanishing damping

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1 The trajectory and the Lyapunov estimate

Let $\mathcal{H}$ be a real Hilbert space. For a maximally monotone operator M, write $J_{\lambda M} = (\mathrm{Id} + \lambda M)^{-1}$. Let $A, B : \mathcal{H} \rightrightarrows \mathcal{H}$ be maximally monotone and let $D : \mathcal{H} \to \mathcal{H}$ be monotone and L-Lipschitz, where $L \geq 0$. Assume that

$$C = \operatorname{zer} B \neq \varnothing, \qquad \Phi = A + D + N_C \text{ is maximally monotone}, \qquad S = \operatorname{zer} \Phi \neq \varnothing. \quad (1)$$

Here N_C is the convex normal cone, with

$$p \in N_C(u) \quad \Longleftrightarrow \quad u \in C \text{ and } \langle p, z - u \rangle \leq 0 \quad (z \in C).$$

The zero sets in (1) are closed and convex. We denote their selected point by $x^* = P_S(0)$.

Fix a constant $0 \le \alpha < 1/3$. The actual iteration is

$$\begin{aligned} &x_0 = x_1 \in \mathcal{H}, \qquad d_n = x_n - x_{n-1}, \\ &y_n = x_n + \alpha d_n, \\ &x_{n+1} = J_{\lambda_n(A+\beta_n B)}\big(y_n - \lambda_n[D(y_n) + \epsilon_n y_n]\big), \qquad n \ge 1. \end{aligned} \tag{2}$$

Assume that $A + \beta_n B$ is maximally monotone for every n. Thus each step uses the resolvent of this sum, and it is well defined even when the initial points lie outside the operator domains.

We retain the following schedules. The sequences $\lambda_n, \beta_n, \epsilon_n$ are positive,

$$\epsilon_n \downarrow 0, \qquad \beta_n \uparrow \infty, \qquad \sum_{n\ge1} \lambda_n^2 < \infty, \qquad \sum_{n\ge1} \lambda_n = \infty. \tag{3}$$

There is a fixed constant c such that

$$0 < c < \frac{1-3\alpha}{2(1+\alpha^2)}, \qquad \lambda_n \le \frac{c\epsilon_n}{2(L+\epsilon_n)^2} \quad (n \ge 1). \tag{4}$$

Put $t_n = \lambda_n \epsilon_n$. We assume that t_n is nonincreasing and that

$$\sum_{n\ge1} \frac{(\epsilon_n - \epsilon_{n+1})^2}{\lambda_{n+1}\epsilon_{n+1}^3} < \infty. \tag{5}$$

There is no upper bound on $\lambda_n \beta_n$ in these assumptions.

Define the constrained Tikhonov path by

$$0 \in \Phi(u_n) + \epsilon_n u_n. \tag{6}$$

Choose graph elements satisfying

$$a_n \in A(u_n), \qquad p_n \in N_C(u_n), \qquad a_n + D(u_n) + \epsilon_n u_n + p_n = 0. \tag{7}$$

For a monotone operator B, its Fitzpatrick function and the support function of C are

$$\begin{aligned} \varphi_B(u,q) &= \sup_{(z,b)\in \operatorname{gra} B} \{\langle u, b\rangle + \langle z, q\rangle - \langle z, b\rangle\}, \\ \sigma_C(q) &= \sup_{z\in C} \langle z, q\rangle. \end{aligned} \tag{8}$$

The penalty assumption is precisely

$$G_n = \varphi_B(u_n, p_n/\beta_n) - \sigma_C(p_n/\beta_n), \qquad \sum_{n\ge1} \lambda_n \beta_n G_n < \infty. \tag{9}$$

This condition refers to the selections in (7). Their norms need not be bounded. Since $p_n \in N_C(u_n)$ and $0 \in B(u_n)$, the support term equals $\langle u_n, p_n/\beta_n\rangle$ and $G_n \ge 0$.

The following theorem centers the conclusion on the energy of the actual trajectory.

Theorem 1. *Under* (1)–(9) *and the stated maximality of the sum resolvents, put*

$$\tau = 1 - 3\alpha, \qquad b_\alpha = \frac{1+5\alpha}{4}, \tag{10}$$

and define

$$\begin{aligned} V_n &= \|x_n - u_n\|^2 - \alpha\|x_{n-1} - u_n\|^2 + b_\alpha\|d_n\|^2, \\ R_n &= 3\lambda_n\beta_n G_n + \frac{3}{\tau t_n}\|u_{n+1} - u_n\|^2. \end{aligned} \tag{11}$$

Then $R_n \geq 0$, $\sum_n R_n < \infty$, *and for every* $n \geq 1$,

$$V_n \geq \frac{4}{45}\big(\|x_n - u_n\|^2 + \|d_n\|^2\big), \tag{12}$$

$$V_{n+1} - V_n + \frac{\tau}{4}\|d_{n+1}\|^2 + \frac{\tau}{2}t_n V_n \leq R_n. \tag{13}$$

In particular, the finite nonnegative Lyapunov function

$$\mathcal{L}_n = V_n + \sum_{k=n}^{\infty} R_k \tag{14}$$

satisfies

$$\mathcal{L}_{n+1} - \mathcal{L}_n \leq -\frac{\tau}{4}\|d_{n+1}\|^2 - \frac{\tau}{2}t_n V_n. \tag{15}$$

The actual sequences obey

$$\sum_{n\geq 1}\|d_{n+1}\|^2 < \infty, \qquad \sum_{n\geq 1} t_n V_n < \infty, \qquad x_n \longrightarrow x^*, \quad y_n \longrightarrow x^* \quad \textit{strongly.} \tag{16}$$

No selected-point interiority or prior boundedness of the orbit is required.

The tail in (14) is a proof quantity. It adds no step to (2). The theorem establishes boundedness and strong selection of the trajectory; it does not give a convergence rate or a finite-length assertion.

The recurrence is the constant-inertia, pure-backward branch of Qu, Staudigl and Peypouquet [1]. Their Theorem 3.4 assumes $x^* \in \operatorname{int}(\operatorname{dom} A)$, which is used to bound the normal selections along (6). The estimate below keeps the normal paired with the implicitly computed point. It consequently uses only the original gap in (9), with no separate term $\lambda_n^2\|p_n\|^2$. All the displayed schedules are retained. In the pure-backward branch the source interprets the forward-penalty cocoercivity constant as infinity, so its corresponding product restriction imposes no finite bound on $\lambda_n\beta_n$.

Implicit normal pairing is already present in the prox-penalization analysis of Attouch, Czarnecki and Peypouquet [2, Lemma 2.1] and in inertial penalty arguments of Balhag, Mazgouri and Théra [3]. The latter strong-convergence results use fixed strong monotonicity. Here the selector vanishes, and the energy follows the moving constrained path. Our comparisons use the indicated public versions. Complete theorem-level comparison

with the final BMT journal text and a related ICOP2024 presentation remains unavailable; no exhaustive priority claim is made. The first result is the self-contained Lyapunov and trajectory theorem above. Section 5 studies a different proximal recurrence whose inertia tends to one. There we use existing value and velocity estimates to construct a trajectory energy at a critical regularization exponent.

2 The constrained reference path

We first record the properties of (6) needed to control its motion.

Lemma 1. *The path is well defined and*

$$\|u_n\| \le \|x^*\|, \qquad u_n \longrightarrow x^* \quad \textit{strongly}, \qquad \|u_{n+1} - u_n\| \le \|x^*\| \frac{\epsilon_n - \epsilon_{n+1}}{\epsilon_{n+1}}. \tag{17}$$

Moreover,

$$0 < t_n \le \frac{1}{4}, \qquad \sum_{n\ge 1} t_n = \infty, \qquad \sum_{n\ge 1} \frac{\|u_{n+1} - u_n\|^2}{t_n} < \infty. \tag{18}$$

Proof. Maximal monotonicity gives $u_n = J_{\epsilon_n^{-1}\Phi}(0)$. Comparing $(u_n, -\epsilon_n u_n)$ with $(x^*, 0)$ yields

$$\|u_n\|^2 \le \langle u_n, x^*\rangle, \qquad \|u_n\| \le \|x^*\|. \tag{19}$$

Every weak cluster point of this bounded path belongs to S: pass to the limit in monotonicity against each fixed point of $\operatorname{gra}\Phi$, using $-\epsilon_n u_n \to 0$, and then use maximality. Its norm is at most $\|x^*\|$, so it is x^*. Weak sequential compactness and uniqueness of the weak cluster point give $u_n \rightharpoonup x^*$. Finally,

$$\|u_n - x^*\|^2 \le \|x^*\|^2 - \langle u_n, x^*\rangle \longrightarrow 0.$$

These arguments include $x^* = 0$, in which case $u_n = 0$ for every n.

Write $h_n = u_{n+1} - u_n$ and $\Delta_n = \epsilon_n - \epsilon_{n+1} \ge 0$. Monotonicity between consecutive reference points gives

$$\epsilon_{n+1}\|h_n\|^2 \le \Delta_n \langle u_n, h_n\rangle.$$

Dividing when $h_n \neq 0$, and using (19), proves the displacement bound in (17).

The step restriction gives

$$t_n \le \frac{c\epsilon_n^2}{2(L+\epsilon_n)^2} \le \frac{c}{2} < \frac{\tau}{4(1+\alpha^2)} \le \frac{1}{4}. \tag{20}$$

To prove the divergent sum, put $r_n = \Delta_n/\epsilon_{n+1}$. Condition (5) says $\sum_n r_n^2/t_{n+1} < \infty$. If $\sum_n t_n < \infty$, Cauchy–Schwarz would give $\sum_n r_n < \infty$. This contradicts

$$\sum_{n=1}^{N} r_n \ge \sum_{n=1}^{N} \log(1 + r_n) = \log \frac{\epsilon_1}{\epsilon_{N+1}} \longrightarrow \infty.$$

Thus the divergent Tikhonov weight is a consequence of the stated assumptions. Since $t_n \geq t_{n+1}$, the path bound also gives

$$\frac{\|h_n\|^2}{t_n} \leq \|x^*\|^2 \frac{\Delta_n^2}{\epsilon_{n+1}^2 t_n} \leq \|x^*\|^2 \frac{\Delta_n^2}{\lambda_{n+1}\epsilon_{n+1}^3}.$$

Summation proves the last assertion. ∎

3 A coercive energy for the actual orbit

The argument keeps the normal selection inside the implicit monotonicity pairing. We first obtain a fixed-center estimate, then account for the change of center.

For $e, d \in \mathcal{H}$, set

$$Q(e,d) = (1-\alpha)\|e\|^2 + 2\alpha\langle e, d\rangle + \frac{1+\alpha}{4}\|d\|^2. \tag{21}$$

The scalar matrix representing this form on $\mathcal{H} \times \mathcal{H}$ has determinant $(1-5\alpha^2)/4 \geq 1/9$ and trace $(5-3\alpha)/4 \leq 5/4$. Its diagonal entries are positive. Its smallest eigenvalue is at least its determinant divided by its trace, and therefore

$$Q(e,d) \geq \frac{4}{45}(\|e\|^2 + \|d\|^2). \tag{22}$$

For an arbitrary center $u \in \mathcal{H}$, define

$$\rho_n(u) = \|x_n - u\|^2 - \alpha\|x_{n-1} - u\|^2 + b_\alpha\|d_n\|^2 = Q(x_n - u, d_n). \tag{23}$$

This proves (12). We will also use an upper bound in terms of the extrapolated point. With $q = y_n - u$ and $d = d_n$, direct substitution of $x_n - u = q - \alpha d$ gives

$$\rho_n(u) = (1-\alpha)\|q\|^2 + 2\alpha^2\langle q, d\rangle + (b_\alpha - \alpha - \alpha^2 - \alpha^3)\|d\|^2.$$

The estimate $2\langle q, d\rangle \leq \|q\|^2 + \|d\|^2$ and the bounds $1 - \alpha + \alpha^2 \leq 1$, $b_\alpha - \alpha - \alpha^3 \leq 1$ imply

$$\rho_n(u) \leq \|y_n - u\|^2 + \|d_n\|^2. \tag{24}$$

Lemma 2. *For each $n \geq 1$, the actual update (2) satisfies*

$$\rho_{n+1}(u_n) + \frac{\tau}{4}\|d_{n+1}\|^2 \leq (1 - \tau t_n)V_n + 2\lambda_n\beta_n G_n. \tag{25}$$

Proof. Fix n and abbreviate $u = u_n$, $p = p_n$, $y = y_n$, $z = x_{n+1}$ and $(\lambda, \beta, \epsilon) = (\lambda_n, \beta_n, \epsilon_n)$. Put $W = D + \epsilon\,\mathrm{Id}$. By the definition of the sum resolvent, there is $b \in B(z)$ such that

$$\frac{y - z}{\lambda} - W(y) - \beta b \in A(z).$$

Monotonicity of A, applied with the selection $a_n = -W(u) - p \in A(u)$, gives

$$\begin{aligned}\|z - u\|^2 \leq \|y - u\|^2 - \|z - y\|^2 - 2\lambda\langle z - u, W(y) - W(u)\rangle \\ + 2\lambda\langle z - u, p - \beta b\rangle.\end{aligned} \tag{26}$$

For the last term, the very graph pair (z, b) belongs to the supremum in (8). Since $\sigma_C(p/\beta) = \langle u, p/\beta\rangle$, this gives

$$\begin{aligned}\langle z-u, p-\beta b\rangle &= \beta\big(\langle u,b\rangle + \langle z, p/\beta\rangle - \langle z,b\rangle - \langle u, p/\beta\rangle\big)\\ &\le \beta\big(\varphi_B(u,p/\beta) - \sigma_C(p/\beta)\big) = \beta G_n.\end{aligned} \tag{27}$$

In particular, this step does not estimate $\|p\|$.

The map W is ϵ-strongly monotone and $(L+\epsilon)$-Lipschitz. Split $z-u = (y-u)+(z-y)$ and apply Young's inequality to the latter term:

$$\begin{aligned}-2\lambda\langle z-u, W(y)-W(u)\rangle \le\ & -2\lambda\epsilon\|y-u\|^2 + c\|z-y\|^2\\ &+ \frac{\lambda^2(L+\epsilon)^2}{c}\|y-u\|^2.\end{aligned} \tag{28}$$

By (4), the last coefficient is at most $\lambda\epsilon/2$. Combining (26)–(28), and discarding an additional nonpositive term, yields

$$\|z-u\|^2 \le \|y-u\|^2 - (1-c)\|z-y\|^2 - t_n\|y-u\|^2 + 2\lambda\beta G_n. \tag{29}$$

We now insert the inertia. The elementary identities and inequality

$$\|y_n-u\|^2 = (1+\alpha)\|x_n-u\|^2 - \alpha\|x_{n-1}-u\|^2 + \alpha(1+\alpha)\|d_n\|^2, \tag{30}$$

$$\|d_{n+1}-\alpha d_n\|^2 \ge (1-\alpha)\|d_{n+1}\|^2 - \alpha(1-\alpha)\|d_n\|^2 \tag{31}$$

show from (29) that

$$\begin{aligned}\rho_{n+1}(u_n) - \rho_n(u_n) \le\ & [b_\alpha - (1-c)(1-\alpha)]\|d_{n+1}\|^2\\ &+ [2\alpha - c\alpha(1-\alpha) - b_\alpha]\|d_n\|^2\\ &- t_n\|y_n-u_n\|^2 + 2\lambda_n\beta_n G_n.\end{aligned} \tag{32}$$

Both square coefficients are at most $-\tau/4$. Indeed,

$$\begin{aligned}2\alpha - c\alpha(1-\alpha) - b_\alpha &= -\frac{\tau}{4} - c\alpha(1-\alpha) \le -\frac{\tau}{4},\\ b_\alpha - (1-c)(1-\alpha) &= -\frac{3\tau}{4} + c(1-\alpha) \le -\frac{\tau}{4},\end{aligned}$$

since $c(1-\alpha) \le \tau/2$. Moreover, $t_n \le 1/4$ and $0 < \tau \le 1$ imply, by (24),

$$\frac{\tau}{4}\|d_n\|^2 + t_n\|y_n-u_n\|^2 \ge \tau t_n\big(\|d_n\|^2 + \|y_n-u_n\|^2\big) \ge \tau t_n V_n. \tag{33}$$

Substituting these estimates into (32) proves (25). ∎

The positive quadratic form also controls the change of center. For any $r, v, h \in \mathcal{H}$ and $\eta > 0$, Young's inequality in the norm induced by Q gives

$$Q(r-h, v) \le (1+\eta)Q(r,v) + (1+\eta^{-1})(1-\alpha)\|h\|^2. \tag{34}$$

Take $r = x_{n+1} - u_n$, $v = d_{n+1}$, $h = u_{n+1} - u_n$ and $\eta = \tau t_n/2 \le 1/8$. The elementary bounds

$$\begin{aligned} (1+\eta)(1-\tau t_n) &\le 1 - \tfrac{1}{2}\tau t_n, \\ 2(1+\eta) &\le 3, \\ (1+\eta^{-1})(1-\alpha) &\le \frac{3}{\tau t_n} \end{aligned} \tag{35}$$

then give, using (25) and weakening the coefficient $(1+\eta)\tau/4$ to $\tau/4$,

$$V_{n+1} + \frac{\tau}{4}\|d_{n+1}\|^2 \le \left(1 - \frac{\tau}{2}t_n\right) V_n + 3\lambda_n\beta_n G_n + \frac{3}{\tau t_n}\|u_{n+1} - u_n\|^2.$$

This is exactly (13). Every term refers to the actual orbit and to the reference path fixed by (6).

4 Dissipation and strong trajectory selection

Completion of the proof of Theorem 1. The nonnegativity and summability of R_n follow from (9) and Lemma 1. Thus the tail in (14) is finite. Subtracting successive tails in (13) gives (15). The coercivity estimate makes $\mathcal{L}_n$ nonnegative, so it has a finite limit. Telescoping yields the explicit dissipation bound

$$\frac{\tau}{4}\sum_{n=1}^{\infty}\|d_{n+1}\|^2 + \frac{\tau}{2}\sum_{n=1}^{\infty} t_n V_n \le \mathcal{L}_1. \tag{36}$$

The error tail tends to zero, and consequently V_n has a limit. Since $V_n \ge 0$, $\sum_n t_n = \infty$, and $\sum_n t_n V_n < \infty$, that limit must be zero. Estimate (12) now implies

$$\|x_n - u_n\| \longrightarrow 0, \qquad \|d_n\| \longrightarrow 0.$$

Together with $u_n \to x^*$ this proves $x_n \to x^*$ and $y_n = x_n + \alpha d_n \to x^*$ strongly. ∎

The energy also supplies an a priori bound for the realized trajectory:

$$\|x_n\| \le \|x^*\| + \sqrt{\frac{45}{4}\mathcal{L}_1} \qquad (n \ge 1). \tag{37}$$

Thus boundedness is obtained before it is used in any conclusion about the orbit. The comparison path is bounded by monotonicity, whereas the actual orbit is bounded by its own coercive energy.

The proof includes $\alpha = 0$, $L = 0$, plateaus in the regularization schedule and $x^* = 0$. In the last case the path is identically zero and its motion contributes nothing to R_n. No division by a normal norm is used, and zero path displacements are treated separately. The conclusion is restricted to constant $\alpha < 1/3$ and the sum resolvent in (2); it does not assert an endpoint or nonconstant-inertia extension.

5 Critical selection with vanishing damping

We now consider a different trajectory, with inertia tending to one. Let $f : \mathbb{R}^d \to \mathbb{R} \cup \{+\infty\}$, $d \geq 1$, be proper, lower-semicontinuous and convex, and assume that $S_f = \operatorname{argmin} f$ is nonempty. Write $z = P_{S_f}(0)$ and $f_* = \min f$. We impose no coercivity or boundedness of S_f. In particular, the minimizers may form an unbounded, nonisolated set. The parameters $\gamma, \kappa, \lambda > 0$ and $0 < q < 1$ are independent of those in Sections 1–4. Starting from arbitrary $x_0, x_1 \in \mathbb{R}^d$, define

$$\begin{aligned} \theta_n &= 1 - \frac{\gamma}{n^q}, \qquad b_n = \frac{\kappa}{n^{q+1}}, \qquad v_n = x_n - x_{n-1}, \\ x_{n+1} &= \operatorname{prox}_{\lambda f}(x_n + \theta_n v_n - b_n x_n), \qquad n \geq 1. \end{aligned} \tag{38}$$

The proximal objective is bounded below by $f_* + \|x - y\|^2/(2\lambda)$ at each finite input y. It therefore attains a unique minimum. Thus every original iterate exists, even through a finite prefix with negative θ_n, and $x_n \in \operatorname{dom} f$ for $n \geq 2$. The initial points need not belong to $\operatorname{dom} f$.

This is the constant-step PIATR recurrence of László [4], at the critical exponent $p = q + 1$. The regularization acts on x_n before the proximal step. The source proves minimum-norm selection for $1 < p < q + 1$, but supplies value and velocity estimates without an orbit-convergence conclusion at equality. We use precisely those established estimates: for every $s \in (1/2, (q+1)/2)$,

$$\begin{aligned} f(x_n) - f_* &= o(n^{-2s}), \qquad \|v_n\| = o(n^{-s}), \\ \sum_{n \geq 1} n^{2s-q} \|v_n\|^2 &< \infty. \end{aligned} \tag{39}$$

These are Theorem 2.1(ii) of [4], with its step exponent set to zero. They hold for every positive γ, κ, λ when $q < 1$, without coercivity or an assumed bounded orbit. Only the asymptotic value estimate and the velocity estimates are imported here. If $f(x_1) = +\infty$, the source's eventual energy proof starts at a sufficiently large original index at least three, where its initial objective term is finite. An objective-value sum printed from index one is then read only on that proximal tail; no such sum is used below. The actual recurrence is not restarted.

Choose an integer $N \geq 2$ such that

$$\frac{\gamma}{N^q} \leq \frac{1}{2}, \qquad N^q \geq 2, \qquad \frac{2q}{\gamma} N^{q-1} \leq \frac{1}{2}, \qquad \frac{\kappa[1 + (2/\gamma)2^q]}{N} \leq \frac{1}{2}. \tag{40}$$

Such an integer exists for every parameter choice under consideration. For $n \geq N$, define

$$r_n = \sum_{j=n}^{\infty} \prod_{i=n}^{j} \theta_i, \qquad \omega_n = (1 + r_{n+1}) b_n, \qquad h_n = \|x_n - z\|^2, \quad \eta_n = r_n (h_n - h_{n-1}). \tag{41}$$

We first introduce an energy whose error terms contain only velocities:

$$\begin{aligned}
\zeta_n &= (1+r_{n+1})\big(2\|v_n\|^2 + b_n\|v_{n+1}\|^2\big), \qquad T_n = \sum_{j=n}^{\infty} \zeta_j, \\
K_n &= (r_n + 2r_n^2)\|v_n\|^2, \qquad U_n = \sup_{j\ge n} K_j, \\
Z_n^{\mathrm{b}} &= h_n + \eta_n + T_n, \qquad m_n = \frac{3}{2}\|z\|^2 + T_n + U_n, \\
A_n &= \max\{Z_n^{\mathrm{b}}, m_n\}, \qquad \mathcal{B}_n = A_n + U_n.
\end{aligned} \tag{42}$$

The first stage of the proof establishes finiteness of these quantities and then derives boundedness. The second stage uses the selection energy

$$\begin{aligned}
\xi_n &= (\|z\|^2 - \|x_n\|^2)_+, \\
e_n &= 2\|v_n\|^2 + 2b_n\|x_n\|\|v_{n+1}\|, \qquad \widehat{e}_n = (1+r_{n+1})e_n, \\
E_n &= \sum_{k=n}^{\infty} \widehat{e}_k, \qquad Z_n = h_n + \eta_n + E_n, \\
\delta_n &= \sup_{j\ge n}(\xi_j + |\eta_j| + E_j), \qquad \mathcal{M}_n = \max\{Z_n, \delta_n\}.
\end{aligned} \tag{43}$$

Here $a_+ = \max\{a, 0\}$. All tails and envelopes are proof quantities along the unchanged trajectory. No online computability or finite-history state representation is asserted. Their finiteness is proved in the stated order, without assuming the selected limit.

Theorem 2. *For the class, initialization and parameters in* (38)*, all quantities in* (41)–(43) *are finite. For every $n \ge N$, the boundedness energy satisfies*

$$\mathcal{B}_n \ge \frac{1}{4}\|x_n - z\|^2 + (r_n + 2r_n^2)\|v_n\|^2 \ge 0, \tag{44}$$

$$\mathcal{B}_{n+1} - \mathcal{B}_n \le -\frac{2}{3}\omega_n(A_n - m_n) - (U_n - U_{n+1}) \le 0. \tag{45}$$

Consequently the whole original orbit is bounded, with

$$\|x_n - z\| \le 2\sqrt{\mathcal{B}_N} \qquad (n \ge N). \tag{46}$$

Moreover, $\delta_n \downarrow 0$, $\sum_{n\ge N}\omega_n = \infty$, and

$$\mathcal{M}_n \ge \frac{1}{4}\|x_n - z\|^2 + (1+r_{n+1})\|v_n\|^2 \ge 0, \tag{47}$$

$$\mathcal{M}_{n+1} - \mathcal{M}_n \le -\omega_n(\mathcal{M}_n - \delta_n) \le 0. \tag{48}$$

In particular,

$$\begin{aligned}
&\mathcal{M}_n \longrightarrow 0, \qquad x_n \longrightarrow z, \\
&\sum_{n=N}^{\infty} \omega_n(\mathcal{M}_n - \delta_n) \le \mathcal{M}_N, \qquad \frac{2}{3}\sum_{n=N}^{\infty} \omega_n(A_n - m_n) \le \mathcal{B}_N.
\end{aligned} \tag{49}$$

Thus the actual trajectory selects the minimum-norm minimizer without a coercivity or bounded-minimizer-set hypothesis.

The weights $1+r_n$ are exactly the classical inertial weights of Attouch, Cabot, Chbani and Riahi [5, Lemma 2.2], following Attouch and Cabot. The final scalar convergence step is also a case of Xu's lemma [6, Lemma 2.5]. The point here is that the signed regularizer provides a position-independent error estimate for a coercive trajectory energy. It establishes boundedness before compactness is used for selection. The value and velocity estimates remain the prior results (39). The theorem is restricted to finite dimensions, constant proximal step and $0 < q < 1$. Our source comparisons use the indicated primary versions; complete final-text coverage of all nearby methods is not claimed.

6 Boundedness before selection

We prove Theorem 2 by justifying the first energy from the velocity estimates alone. Only after its coercivity bounds the original phase will we use compactness to construct the selection energy.

6.1 Weights and position-independent tails

For all $n \geq N$, condition (40) gives $1/2 \leq \theta_n < 1$. Truncate the series in (41) at $m \geq n$, writing $r_n^{[m]}$ and setting $r_{m+1}^{[m]} = 0$. The exact finite recursion is $r_n^{[m]} = \theta_n(1 + r_{n+1}^{[m]})$. The sequence $W_n = (2/\gamma)n^q$ is a supersolution: by concavity of u^q,

$$\begin{aligned} W_n - \theta_n(1 + W_{n+1}) &= \frac{2}{\gamma}[n^q - (n+1)^q] + 2\left(\frac{n+1}{n}\right)^q - \theta_n \\ &\geq -\frac{2q}{\gamma}n^{q-1} + 2 - \theta_n \geq \frac{1}{2}. \end{aligned} \tag{50}$$

Backward induction bounds $r_n^{[m]}$ by W_n. Passing to the increasing limit in m proves finiteness and the exact identity

$$r_n = \theta_n(1 + r_{n+1}), \qquad r_n \leq \frac{2}{\gamma}n^q. \tag{51}$$

For a lower bound retain the first $\lfloor n^q \rfloor$ products. Each is at least $e^{-2\gamma}$, because $\log(1-u) \geq -2u$ for $0 \leq u \leq 1/2$ and its length is at most n^q. Since $\lfloor n^q \rfloor \geq n^q/2$, we obtain

$$\frac{e^{-2\gamma}}{2}n^q \leq r_n \leq \frac{2}{\gamma}n^q, \qquad \frac{\kappa e^{-2\gamma}}{2n} \leq \omega_n \leq \frac{\kappa[1 + (2/\gamma)2^q]}{n} \leq \frac{1}{2}. \tag{52}$$

In particular, $\sum_n \omega_n = \infty$.

Choose

$$\max\{q, 1/2\} < s < (q+1)/2. \tag{53}$$

This interval is nonempty for every $0 < q < 1$. From (39),

$$f(x_n) \longrightarrow f_*, \qquad n^q\|v_n\| \longrightarrow 0, \qquad \sum_n n^q\|v_n\|^2 < \infty. \tag{54}$$

The last implication uses $2s - q > q$. Set $D = 1 + (2/\gamma)2^q$. The first velocity error obeys

$$0 \le \zeta_n \le 2Dn^q\|v_n\|^2 + \kappa Dn^{-1}\|v_{n+1}\|^2.$$

Both terms are summable by (54), including the shifted velocity sum. Hence T_n is finite, $T_n \downarrow 0$, and $T_n - T_{n+1} = \zeta_n$. Also,

$$0 \le K_n \le \left(\frac{2}{\gamma} + \frac{8}{\gamma^2}\right) n^{2q}\|v_n\|^2 \longrightarrow 0.$$

It follows that U_n is finite, nonincreasing and tends to zero. This uses no attainment of the tail supremum. Thus every quantity in (42) is finite without a bound on the positions. The tails tend to zero; no summability of the sequence (U_n) or of the sequence (T_n) is assumed.

6.2 The actual proximal step and phase control

Proximal optimality gives

$$g_{n+1} = \lambda^{-1}(\theta_n v_n - v_{n+1} - b_n x_n) \in \partial f(x_{n+1}), \qquad \langle g_{n+1}, x_{n+1} - z\rangle \ge f(x_{n+1}) - f_* \ge 0.$$

Writing $\Delta h_n = h_n - h_{n-1}$, polarization yields the exact identity

$$\begin{aligned}\Delta h_{n+1} - \theta_n \Delta h_n = \theta_n(1+\theta_n)\|v_n\|^2 - \|v_{n+1} - \theta_n v_n\|^2 \\ - 2b_n\langle x_n, x_{n+1} - z\rangle - 2\lambda\langle g_{n+1}, x_{n+1} - z\rangle.\end{aligned} \tag{55}$$

The regularizer has remained at its literal point x_n. Its norm identity

$$-2\langle x_n, x_{n+1} - z\rangle = -h_n + \|z\|^2 + \|v_{n+1}\|^2 - \|x_{n+1}\|^2 \tag{56}$$

retains the sign needed for boundedness. Since $0 \le \theta_n \le 1$, dropping the nonpositive terms in (55) gives

$$\Delta h_{n+1} - \theta_n \Delta h_n \le -b_n(h_n - \|z\|^2) + 2\|v_n\|^2 + b_n\|v_{n+1}\|^2. \tag{57}$$

In particular, the right-hand error contains no uncontrolled position norm.

The Green identity cancels the inertial distance difference exactly:

$$(h_{n+1} + \eta_{n+1}) - (h_n + \eta_n) = (1 + r_{n+1})(\Delta h_{n+1} - \theta_n \Delta h_n). \tag{58}$$

Multiplying (57) by $1 + r_{n+1}$ and subtracting the velocity-error tail yields

$$Z^{\mathrm{b}}_{n+1} - Z^{\mathrm{b}}_n \le -\omega_n h_n + \omega_n\|z\|^2. \tag{59}$$

To handle the indefinite history term, observe the square identities

$$h_n + \eta_n - \frac{1}{2}h_n + K_n = \frac{1}{2}\|x_n - z + 2r_n v_n\|^2, \tag{60}$$

$$\frac{3}{2}h_n + K_n - (h_n + \eta_n) = \frac{1}{2}\|x_n - z - 2r_n v_n\|^2 + 2r_n\|v_n\|^2. \tag{61}$$

They give

$$\frac{1}{2}h_n - K_n \le h_n + \eta_n \le \frac{3}{2}h_n + K_n. \tag{62}$$

Since $K_n \le U_n$, the upper bound implies $h_n \ge (2/3)(Z_n^{\mathrm{b}} - T_n - U_n)$. Thus (59) becomes

$$Z_{n+1}^{\mathrm{b}} \le \left(1 - \frac{2}{3}\omega_n\right) Z_n^{\mathrm{b}} + \frac{2}{3}\omega_n m_n. \tag{63}$$

We use a simple maximum comparison. If $0 \le a \le 1$, $F' \le (1-a)F + aC$ and $C' \le C$, then

$$\max\{F', C'\} - \max\{F, C\} \le -a\big(\max\{F, C\} - C\big). \tag{64}$$

For $F \ge C$, the number $(1-a)F + aC$ bounds both new entries, since it is at least $C \ge C'$. For $F \le C$, both new entries are at most C. These two arguments include ties and either choice of the new maximum.

Here $m_{n+1} \le m_n$, because T_n and U_n are nonincreasing. Apply (64) to (63), then add $U_{n+1} - U_n$. This proves (45). For coercivity, the lower bound in (62) gives

$$\mathcal{B}_n \ge Z_n^{\mathrm{b}} + U_n \ge \frac{1}{2}h_n - K_n + T_n + U_n \ge \frac{1}{2}h_n.$$

The other maximum branch gives $\mathcal{B}_n \ge m_n + U_n \ge 2U_n \ge 2K_n$. Averaging these lower bounds proves (44). Since $\mathcal{B}_N$ is already known to be finite, descent and coercivity give (46). The finite original prefix is bounded as well. This establishes the actual orbit bound before invoking compactness. Telescoping (45) also proves the last sum estimate in (49).

6.3 The selection energy after boundedness

Every cluster point of the now bounded orbit is a minimizer, by lower semicontinuity and (54). If ξ_n failed to tend to zero, a convergent subsequence would yield a point of S_f with norm strictly less than $\|z\|$. Therefore

$$\xi_n \longrightarrow 0. \tag{65}$$

Only the orbit has been made bounded; compactness of S_f is not needed. This argument does not identify any cluster point with z.

Boundedness and (52) now imply

$$|\eta_n| \le r_n(\|x_n - z\| + \|x_{n-1} - z\|)\|v_n\| \longrightarrow 0.$$

The first term of $\widehat{e}_n$ is summable by (54). For its second term, boundedness reduces the question to $\sum_n \|v_{n+1}\|/n < \infty$. For example, Cauchy–Schwarz gives

$$\sum_{n\ge N} \frac{\|v_{n+1}\|}{n} \le \left(\sum_{n\ge N} (n+1)^q \|v_{n+1}\|^2\right)^{1/2} \left(\sum_{n\ge N} \frac{1}{n^2(n+1)^q}\right)^{1/2} < \infty. \tag{66}$$

Hence E_n is finite, decreases to zero, and $E_{n+1} - E_n = -\widehat{e}_n$. Since $\xi_n + |\eta_n| + E_n \to 0$, its tail supremum δ_n is finite, nonnegative and decreases to zero. Every quantity in (43) has thus been justified before selection.

Return to (55), this time using

$$2\langle x_n, x_{n+1} - z\rangle = h_n + \|x_n\|^2 - \|z\|^2 + 2\langle x_n, v_{n+1}\rangle.$$

Bound only the last pairing in absolute value. The result is

$$\Delta h_{n+1} - \theta_n \Delta h_n + b_n h_n \le b_n \xi_n + e_n. \tag{67}$$

Multiply by $1 + r_{n+1}$, use (58), and add the selection-tail difference. We obtain

$$Z_{n+1} - Z_n \le -\omega_n h_n + \omega_n \xi_n \le -\omega_n Z_n + \omega_n \delta_n.$$

The last step uses $h_n = Z_n - \eta_n - E_n$ and $\xi_n + \eta_n + E_n \le \delta_n$. Applying (64), with the nonincreasing envelope δ_n, proves (48).

For coercivity, $\mathcal{M}_n$ bounds both $Z_n = h_n + \eta_n + E_n$ and $|\eta_n| + E_n \le \delta_n$. Averaging gives $\mathcal{M}_n \ge h_n/2 + E_n$. Also $E_n \ge \widehat{e}_n \ge 2(1 + r_{n+1})\|v_n\|^2$. These estimates imply the conservative bound (47).

Finally, the nonnegative decreasing sequence $\mathcal{M}_n$ has a limit $\ell \ge 0$. Telescoping (48) gives

$$\sum_{n=N}^{m} \omega_n(\mathcal{M}_n - \delta_n) \le \mathcal{M}_N.$$

If $\ell > 0$, eventually $\delta_n \le \ell/2$, contradicting the divergence of $\sum_n \omega_n$. Thus $\mathcal{M}_n \to 0$, and coercivity gives $x_n \to z$. Letting $m \to \infty$ proves the remaining dissipation assertion in (49) and completes the proof of Theorem 2.

The finite initial interval remains part of the trajectory throughout. Only the energy estimates start at N, determined by (40). No initial-domain or equal-history condition has been imposed. The case $z = 0$ and vanishing tails are included, with no division by any state or energy norm. The result gives neither a norm-convergence rate nor a finite-length statement; the value and velocity estimates remain those of [4].

7 An explicit trajectory with gradient history

We now consider a third recurrence, independently of the two preceding schemes. Let $f : \mathbb{R}^d \to \mathbb{R}$ be convex and C^2, where $d < \infty$, and suppose that ∇f is globally 1-Lipschitz and $S = \operatorname{argmin} f \ne \varnothing$. Write $f_* = \min f$, $z = P_S(0)$, and $Z = \|z\|$. Strong convexity of f, coercivity, and boundedness of S are not assumed. From arbitrary $x_0, x_1 \in \mathbb{R}^d$, define, for every $k \ge 1$,

$$\begin{aligned} s = \frac{1}{4}, \qquad \mu_k = \frac{1}{k}, \qquad a_k = \frac{3}{2\sqrt{k}}, \qquad d_k = x_k - x_{k-1}, \\ y_k = x_k + (1 - a_k)d_k - s\big(\nabla f(x_k) - \nabla f(x_{k-1})\big), \\ x_{k+1} = y_k - s\big(\nabla f(y_k) + \mu_k y_k\big). \end{aligned} \tag{68}$$

This is the named IGATH-2 scheme of Attouch, Balhag, Chbani and Riahi [7, Section 4.2], with $\varepsilon_k = 1/k$, $\delta = 3$, $s = 1/4$, and $\beta = 1/2$. The method is inherited. We give a self-contained energy estimate for this fixed specialization, without asserting a maximal parameter range. The gradient difference in (68) is unregularized, and the regularizer acts at y_k. Every step is an explicit full-domain operation. In particular, the negative inertia coefficients at $k = 1, 2$ cause no solvability problem and are retained in the trajectory.

Set $\tau_k = 1/(64\sqrt{k})$. The first phase functional is

$$\begin{aligned}\mathcal{H}_k(x,p) = \frac{k}{4}(f(x) - f_*) + \frac{1}{8}\|x\|^2 + \frac{k}{2}\|x-p\|^2 \\ + \frac{\sqrt{k}}{64}\langle p - z, x - p + s\nabla f(p)\rangle + \frac{3}{256}\|p-z\|^2, \qquad H_k = \mathcal{H}_k(x_k, x_{k-1}).\end{aligned} \tag{69}$$

For the second functional, let

$$\phi_k(x) = f(x) + \frac{\mu_k}{2}\|x\|^2, \qquad z_k = \operatorname{argmin}\phi_k, \qquad A_k = a_k - s\mu_k. \tag{70}$$

Each z_k exists uniquely because ϕ_k is strongly convex and is bounded below by $f_* + \mu_k\|x\|^2/2$. Define

$$\begin{aligned}M_k(x,p) &= s\big(\phi_k(x) - \phi_k(z_k)\big) + \frac{1}{2}\|x-p\|^2, \\ \mathcal{E}_k(x,p) &= M_k(x,p) + \tau_k\langle p - z_k, x - p + s\nabla\phi_k(p)\rangle + \frac{\tau_k A_k}{2}\|p - z_k\|^2, \\ E_k &= \mathcal{E}_k(x_k, x_{k-1}).\end{aligned} \tag{71}$$

The regularized gradient in this energy does not change the gradient history in the algorithm.

Theorem 3. *For the class and original initialization above, the following assertions hold.*

(i) *For every phase pair (x,p) and every $k \geq 1$,*

$$\mathcal{H}_k(x,p) \geq \frac{k}{4}(f(x) - f_*) + \frac{1}{8}\|x\|^2 + \frac{k}{4}\|x-p\|^2 + \frac{47}{4096}\|p-z\|^2. \tag{72}$$

For $K = 360000$ and every $k \geq K$,

$$H_{k+1} \leq \left(1 - \frac{1}{512\sqrt{k}}\right) H_k + \frac{5Z^2}{512\sqrt{k}}. \tag{73}$$

Thus, with

$$B = \max\{H_1, \ldots, H_K, 5Z^2\}, \qquad X = \sqrt{8B}, \quad D = 2\sqrt{B}, \quad G = \sqrt{8B}, \tag{74}$$

we have $H_k \leq B$ and, for every $k \geq 1$,

$$\|x_k\| \leq X, \qquad \|d_k\| \leq \frac{D}{\sqrt{k}}, \qquad 0 \leq f(x_k) - f_* \leq \frac{4B}{k}, \qquad \|\nabla f(x_k)\| \leq \frac{G}{\sqrt{k}}. \tag{75}$$

(ii) *For every independent phase pair and every $k \geq 4$,*

$$\frac{3}{4}M_k(x,p) \leq \mathcal{E}_k(x,p) \leq 2M_k(x,p), \tag{76}$$

$$\begin{aligned}\mathcal{E}_k(x,p) &\geq s\big(\phi_k(x) - \phi_k(z_k)\big) + \frac{3}{8}\|x-p\|^2 + \frac{\tau_k A_k}{4}\|p-z_k\|^2 \\ &\geq \frac{1}{8k}\|x-z_k\|^2 + \frac{3}{8}\|x-p\|^2.\end{aligned} \tag{77}$$

Put

$$\begin{aligned}&N = 2^{20}, \qquad R = X + Z, \qquad V = D + sG + sX = D + X/2, \qquad a_0 = 1/256, \\ &C_0 = \frac{Z^2}{8} + \frac{RV}{128} + \frac{ZV}{64} + \frac{RX}{256} + \frac{3RZ}{128}.\end{aligned} \tag{78}$$

Along the actual trajectory,

$$E_{k+1} \leq \left(1 - \frac{a_0}{\sqrt{k}}\right) E_k + \frac{C_0}{k^2} \qquad (k \geq N). \tag{79}$$

(iii) *Let $Q = \max\{N^{3/2}E_N, 2C_0/a_0\}$. For $k \geq N$,*

$$\begin{aligned}&E_k \leq \frac{Q}{k^{3/2}}, \qquad \|x_k - z_k\|^2 \leq \frac{8Q}{\sqrt{k}}, \qquad \|d_k\|^2 \leq \frac{8Q}{3k^{3/2}}, \\ &0 \leq f(x_k) - f_* \leq \frac{4Q}{k^{3/2}} + \frac{Z^2}{2k}.\end{aligned} \tag{80}$$

Moreover,

$$x_k \longrightarrow z, \qquad y_k \longrightarrow z, \qquad \sum_{k\geq 1}\|d_k\|^2 < \infty. \tag{81}$$

The first energy is uniformly coercive in the original phase. It proves the bounds used to control the motion of the second energy. The large integers K, N are conservative proof thresholds; they impose no change or restart on (68). The finite maximum in (74) is an analysis constant, not an additional computational step. The norm estimate in (80) concerns the Tikhonov path; no quantitative rate for $z_k \to z$ or finite trajectory length is asserted.

The choice of a previous-position pairing follows the familiar use of mixed momentum and regularized reference paths in [7, 8]. The proof below verifies it for the literal explicit recurrence. TRIGA in [8] omits its gradient-difference term; the classical IGAHD map in [10] contains an additional previous-gradient correction and has a different damping schedule. The implicit algorithm of Sun, Tian and Zhang [9] uses a future-gradient difference and an increasing time scale. Neither its strict selection range nor the fixed-objective Ravine/Nesterov interchange in [11] is a direct instance of the theorem above. Our comparison also inspected the stochastic Ravine author version [12]; its conditional-unbiasedness assumption is not met merely by treating a deterministic state-dependent regularizer as an error. These are exact-method comparisons, not a classification of all

possible transformations. The complete final ABCR article was read as author-posted text. The inspected STZ and stochastic Ravine bodies are the indicated author versions; complete comparison with their journal-final bodies remains unavailable. No firstness or exhaustive priority claim is made.

8 Proof of Theorem 3

8.1 Two identities for the actual step

Convexity and global smoothness give $0 \preceq \nabla^2 f \preceq I$ and

$$\|\nabla f(x)\|^2 \le 2(f(x) - f_*). \tag{82}$$

The latter follows by applying the descent inequality at $x - \nabla f(x)$ and using $f_* \le f(x - \nabla f(x))$.

Fix $k \ge 4$, and abbreviate $t = k^{-1/2}$, $a = 3t/2$, $\mu = t^2$, $\tau = t/64$, $A = a - s\mu$ and $L = 1 + \mu$. For an independent pair (x, p) set $d = x - p$, $u = y - x$, $g_y = \nabla\phi_k(y)$ and $d^+ = x^+ - x = u - sg_y$, where (y, x^+) is the actual step at index k. The average Hessian of f on the segment from p to x lies between zero and I. Since $1 - a \ge s$,

$$u = \big((1-a)I - s\overline{H}\big)d, \qquad 0 \preceq (1-a)I - s\overline{H} \preceq (1-a)I, \qquad \|u\| \le (1-a)\|d\|. \tag{83}$$

In particular, this calculation uses the unregularized gradient history.

For any fixed scalar c, write $M_c(x, p) = s(\phi_k(x) - c) + \|d\|^2/2$. Strong convexity from y to x and smoothness from y to $x^+ = y - sg_y$ give, after cancellation of the mixed terms,

$$\begin{aligned} M_c(x^+, x) - M_c(x, p) &\le \frac{1}{2}(\|u\|^2 - \|d\|^2) - \frac{s^2(1 - sL)}{2}\|g_y\|^2 - \frac{s\mu}{2}\|u\|^2 \\ &\le -(a - a^2/2)\|d\|^2 - \frac{s^2(1 - sL)}{2}\|g_y\|^2 - \frac{s\mu}{2}\|u\|^2. \end{aligned} \tag{84}$$

A second exact identity treats the gradient history. Suppose that $u = (1 - b)d - s(\nabla h(x) - \nabla h(p))$, and keep b and a center c_0 fixed during the step. Set

$$T_{h,b,c_0}(x, p) = \langle p - c_0, d + s\nabla h(p)\rangle + \frac{b}{2}\|p - c_0\|^2.$$

The vector $w = d + s\nabla h(p)$ obeys $d^+ + s\nabla h(x) = w - bd - sg_y$. Expansion therefore yields

$$T_{h,b,c_0}(x^+, x) - T_{h,b,c_0}(x, p) = (1 - b/2)\|d\|^2 + s\langle d, \nabla h(p)\rangle - s\langle x - c_0, g_y\rangle. \tag{85}$$

The previous position p in this definition is essential to the cancellation. We shall use it first with $(h, b, c_0) = (f, a, z)$ and then with (ϕ_k, A, z_k). For any center c_0, the same two function inequalities also give

$$-s\langle x - c_0, g_y\rangle \le -s\big(\phi_k(x) - \phi_k(c_0)\big) + \frac{sL}{2}\|u\|^2 - \frac{s\mu}{2}\|y - c_0\|^2. \tag{86}$$

Indeed, combine strong convexity toward c_0 with $\phi_k(x) \le \phi_k(y) - \langle g_y, u\rangle + L\|u\|^2/2$.

8.2 A fixed minimizer gives boundedness first

The inequality $\langle p - z, \nabla f(p)\rangle \geq 0$ and Young's inequality imply

$$\frac{\sqrt{k}}{64}\langle p - z, d\rangle \geq -\frac{k}{4}\|d\|^2 - \frac{1}{4096}\|p - z\|^2.$$

This proves (72) for every $k \geq 1$, including the original negative-inertia prefix.

For $k \geq 4$, put

$$P = \frac{k}{4}(f(x) - f_*) + \frac{1}{8}\|x\|^2 + \frac{k}{2}\|d\|^2.$$

We shall use the global upper comparison

$$\mathcal{H}_k(x, p) \leq 2P + Z^2/16. \tag{87}$$

Here are explicit coefficients for this estimate. Set $U = \|p - z\|$, $e = \sqrt{k}\|d\|$ and $r = \sqrt{k(f(x) - f_*)}$. By (82) and Lipschitz continuity,

$$\mathcal{H}_k - P \leq \frac{5}{256}Ue + \frac{\sqrt{2}}{256}Ur + \frac{3}{256}U^2 \leq \frac{13}{512}U^2 + \frac{5}{512}e^2 + \frac{1}{512}r^2.$$

Since $U^2 \leq 4\|x\|^2 + e^2 + 2Z^2$, the right side is at most

$$\frac{13}{128}\|x\|^2 + \frac{18}{512}e^2 + \frac{1}{512}r^2 + \frac{13}{256}Z^2 \leq P + Z^2/16.$$

No trajectory bound was used.

Apply (85) with (f, a, z). The gradient-gap bound implies

$$s\langle d, \nabla f(p)\rangle \leq \frac{s}{2}(f(x) - f_*) + 2s\|d\|^2.$$

Using (86), $\phi_k(z) = f_* + \mu Z^2/2$ and (83), we obtain

$$\Delta T_{f,a,z} \leq 2\|d\|^2 - \frac{s}{2}(f(x) - f_*) - \frac{s\mu}{2}\|x\|^2 + \frac{s\mu}{2}Z^2. \tag{88}$$

For example the velocity coefficient before rounding is $1 - a/2 + 2s + sL/2 \leq 53/32 < 2$. The elementary bounds

$$a - a^2/2 - 2\tau \geq a/2 + \tau/4, \qquad (1 - sL)/2 \geq 1/4$$

then combine (84) and (88) into

$$\mathcal{H}_k(x^+, x) \leq (1 - \tau/4)\mathcal{H}_k(x, p) - \frac{ka}{2}\|d\|^2 - \frac{ks^2}{4}\|g_y\|^2 + \frac{9\tau}{64}Z^2. \tag{89}$$

In this step one first obtains the loss $\tau P/2$ and the forcing $\tau Z^2/8$, and then uses (87).

The actual weight change must still be paid. Let $c_k = 1 + 1/k$ and $\delta_k = ((k+1)/\sqrt{k} - \sqrt{k+1})/64$. Then $0 \leq \delta_k \leq 1/(64\sqrt{k})$, and direct subtraction gives

$$\begin{aligned}\mathcal{H}_{k+1}(x^+, x) = c_k\mathcal{H}_k(x^+, x) - \frac{\|x^+\|^2}{8k} \\ - \delta_k\langle x - z, d^+ + s\nabla f(x)\rangle - \frac{3}{256k}\|x - z\|^2.\end{aligned} \tag{90}$$

The gradient pairing in its second line is nonpositive. Completing a square in $x - z$ bounds the remaining two terms by $\|d^+\|^2/192$. Since $\|d^+\|^2 \le 2\|d\|^2 + 2s^2\|g_y\|^2$, the losses retained in (89) absorb this quantity. We conclude that

$$H_{k+1} \le c_k(1 - \tau_k/4)H_k + c_k(9\tau_k/64)Z^2.$$

For $k \ge K$, $\sqrt{k} \ge 600 \ge 512$ gives $1/k \le \tau_k/8$, while $c_k \le 5/4$. These inequalities imply (73). Its coefficient is a convex weight, so induction from the actual H_K proves $H_k \le B$. Coercivity and (82) now give (75). Also $x_0 = x_1 - d_1$ is bounded by $X + D$. Thus boundedness of the original orbit has been established before any moving-center estimate is invoked.

8.3 The second energy on the whole phase

Fix $k \ge 4$, write $F = \phi_k(x) - \phi_k(z_k)$ and $M = sF + \|d\|^2/2$. We have

$$\frac{11}{8}t \le A \le \frac{3}{2}t, \qquad \frac{\tau}{A} \le \frac{1}{88}.$$

Since $\langle p - z_k, \nabla\phi_k(p)\rangle \ge 0$, square completion gives

$$\mathcal{E}_k(x,p) \ge sF + (1/2 - \tau/A)\|d\|^2 + \frac{\tau A}{4}\|p - z_k\|^2.$$

This proves the lower comparisons in (76)– (77). For the upper bound, let $F_p = \phi_k(p) - \phi_k(z_k)$. Strong convexity and smoothness give

$$\mu\|p - z_k\|^2 \le 4F + \tfrac{1}{2}\|d\|^2, \qquad F_p \le 2F + \tfrac{5}{4}\|d\|^2, \qquad \|p - z_k\|\|\nabla\phi_k(p)\| \le 2\sqrt{L/\mu}\,F_p.$$

Consequently the three additions to M in the energy are bounded by

$$\begin{aligned} \tau\,|\langle p - z_k, d\rangle| &\le F/32 + 3\|d\|^2/256, \\ s\tau\langle p - z_k, \nabla\phi_k(p)\rangle &\le F/32 + 5\|d\|^2/256, \\ (\tau A/2)\|p - z_k\|^2 &\le 3F/64 + 3\|d\|^2/512. \end{aligned}$$

Thus $\mathcal{E}_k \le 23F/64 + 275\|d\|^2/512 \le 2M$. All comparisons are global in (x,p).

For the frozen step, the exact identity

$$u = (1 - A)d - s\big(\nabla\phi_k(x) - \nabla\phi_k(p)\big)$$

allows (85) with (ϕ_k, A, z_k). Since $\|p - y\| \le 2\|d\|$, smoothness and Young's inequality yield

$$s\langle d, \nabla\phi_k(p)\rangle \le \tfrac{1}{2}\|d\|^2 + \tfrac{s^2}{2}\|g_y\|^2 + 2sL\|d\|^2.$$

Using (86) shows $\Delta T_{\phi_k,A,z_k} \le 3\|d\|^2 + s^2\|g_y\|^2/2 - sF$. Together with (84), this gives

$$\Delta\mathcal{E}_k \le -(a - a^2/2 - 3\tau)\|d\|^2 - s^2\big((1 - sL)/2 - \tau/2\big)\|g_y\|^2 - \tau sF.$$

The two coefficients are bounded below by $\tau/2$ and $87/256$, respectively. If $E_{k+1}^{[k]}$ denotes the new phase evaluated with the old coefficients and old center, we therefore have

$$E_{k+1}^{[k]} \le E_k - \tau_k M_k \le (1-\tau_k/2)E_k \le (1-\tau_k/4)E_k. \tag{91}$$

This still has a frozen center; it is not yet (79).

8.4 Every change of the center and coefficients

The reference path satisfies

$$\|z_k\| \le Z, \quad 0 \le f(z_k) - f_* \le \frac{Z^2}{2k}, \quad \|z_{k+1} - z_k\| \le \frac{Z}{k+1}, \quad z_k \longrightarrow z. \tag{92}$$

Indeed compare the minimum value of ϕ_k with $\phi_k(z)$. For the displacement use strong monotonicity of $\nabla\phi_k$ and $\nabla\phi_k(z_{k+1}) = (\mu_k - \mu_{k+1})z_{k+1}$. A zero displacement is included without division. The path lies in the ball of radius Z; every cluster point is a minimizer of norm at most Z. Finite-dimensional compactness and uniqueness of the minimum-norm minimizer prove the last limit. This reference-path argument is independent of the actual trajectory.

At the new phase put $p' = x_k$, $x' = x_{k+1}$, $d' = d_{k+1}$, and $r_j = p' - z_j$, $w_j = d' + s\nabla\phi_j(p')$ for $j = k, k+1$. The already established bounds (75) imply

$$\|r_k\|, \|r_{k+1}\| \le R, \qquad \|w_k\| \le V/\sqrt{k}. \tag{93}$$

Write $\Delta\mu_k = 1/[k(k+1)]$. The change of the potential-gap part is at most $s\Delta\mu_k Z^2/2 \le Z^2/(8k^2)$, because

$$\min\phi_k - \min\phi_{k+1} \le \Delta\mu_k \|z_{k+1}\|^2/2$$

and the change at x' has an additional nonpositive term. The kinetic part is unchanged. The exact change of the pairing is

$$\begin{aligned}\tau_{k+1}\langle r_{k+1}, w_{k+1}\rangle - \tau_k\langle r_k, w_k\rangle &= (\tau_{k+1}-\tau_k)\langle r_k, w_k\rangle \\ &\quad + \tau_{k+1}\langle r_{k+1}-r_k, w_k\rangle + \tau_{k+1}\langle r_{k+1}, w_{k+1}-w_k\rangle.\end{aligned} \tag{94}$$

Here $|\tau_{k+1} - \tau_k| \le 1/(128k^{3/2})$ and $w_{k+1} - w_k = -s\Delta\mu_k p'$. Thus the three upper costs are, respectively,

$$\frac{RV}{128k^2}, \qquad \frac{ZV}{64k^2}, \qquad \frac{RX}{256k^2}.$$

Finally,

$$b_k = \tau_k A_k/2 = \frac{3}{256k} - \frac{1}{512k^{3/2}}$$

is positive and decreasing. Its coefficient change is nonpositive, and

$$b_{k+1}\|r_{k+1}\|^2 - b_k\|r_k\|^2 \le b_{k+1}(\|r_{k+1}\| + \|r_k\|)\|z_{k+1} - z_k\| \le \frac{3RZ}{128k^2}.$$

The difference-of-squares formula already includes the entire center shift. Adding these five costs proves $E_{k+1} \le E_{k+1}^{[k]} + C_0/k^2$. Combining with (91) yields (79). Every position or gradient bound used here came from the first energy.

8.5 Selection from the actual energy inequality

The definition of Q gives $E_N \le QN^{-3/2}$ and $C_0 \le a_0 Q/2$. For $k \ge N$, $\sqrt{k} \ge 1024 \ge 3/a_0$ and

$$1 - \frac{a_0}{2\sqrt{k}} \le 1 - \frac{3}{2k} \le \left(\frac{k}{k+1}\right)^{3/2}.$$

The last inequality is the tangent bound for $(1+t)^{-3/2}$. Induction in (79) proves $E_k \le Qk^{-3/2}$, including $Q = 0$. Coercivity gives the two distance and velocity estimates in (80). Also

$$f(x_k) - f_* \le \phi_k(x_k) - \phi_k(z_k) + \frac{Z^2}{2k},$$

which gives the objective estimate. The limits $x_k - z_k \to 0$ and $z_k \to z$ imply $x_k \to z$; (83) then gives $y_k \to z$. The increment bound has summable squares, and the finite original prefix adds only finitely many terms.

If $B = 0$, coercivity gives $z = x_0 = x_1 = 0$, and the actual trajectory is constant. If $Z = 0$ with nonzero initial data, $z_k = 0$ but the remaining terms of C_0 are retained. Constant objectives are also included. No step divides by a data norm or an energy, assumes bounded minimizers, or replaces the original initialization. This completes the proof.

9 A composite trajectory with a moving gradient center

We next retain a genuine nonsmooth term. Let $H : \mathbb{R}^d \to \mathbb{R}$ be convex and C^2, with globally 1-Lipschitz gradient, and let $G : \mathbb{R}^d \to \mathbb{R} \cup \{+\infty\}$ be proper, lower semicontinuous and convex. Assume that $d < \infty$ and

$$F = H + G, \qquad S = \operatorname{argmin} F \ne \varnothing, \qquad F_* = \min F, \qquad z^* = P_S(0), \qquad Z = \|z^*\|.$$

The notation in this section is local to the following fourth recurrence. Starting from arbitrary finite $x_0, x_1 \in \mathbb{R}^d$, set, for every $k \ge 1$,

$$\begin{aligned} &t_k = k^{-1/2}, \quad \mu_k = k^{-1}, \quad a_k = (1 + 3t_k/2)^{-1}, \quad d_k = a_k/4, \quad b_k = 1 - a_k, \quad \tau_k = t_k/64, \\ &v_k = x_k - x_{k-1}, \\ &y_k = x_k + a_k v_k - d_k\big(\nabla H(x_k) - \nabla H(x_{k-1})\big), \\ &x_{k+1} = \operatorname{prox}_{d_k G}\big(y_k - d_k[\nabla H(y_k) + \mu_k x_k]\big). \end{aligned} \tag{95}$$

Here d_k denotes the positive proximal step. The history uses only the smooth gradient, and the regularizer acts at the old point x_k. This is the IPGATH-1 display of [7, Section 5.2], with $s = 1/4$, $\delta = 3$, $\beta = 1/2$ and $\varepsilon_k = 1/k$. That display supplies no separate initialization line; we explicitly use the arbitrary two-point convention in (95). Every

proximal output belongs to $\operatorname{dom} G$, so objective energies are finite from original index 2. Neither $G(x_1)$ nor $G(y_k)$ is required to be finite.

For $x \in \operatorname{dom} G$ and $p \in \mathbb{R}^d$, define the first phase energy by

$$\begin{aligned}\mathcal{H}_k^{\mathrm{c}}(x,p) &= \frac{k}{4}(F(x)-F_*) + \frac{1}{8}\|x\|^2 + \frac{k}{2a_k}\|x-p\|^2 \\ &\quad + \frac{\sqrt{k}}{64a_k}\langle p-z^*, x-p\rangle + \frac{\sqrt{k}}{256}\langle p-z^*, \nabla H(p)-\nabla H(z^*)\rangle + \frac{3}{256}\|p-z^*\|^2 \qquad (96)\\ H_k^{\mathrm{c}} &= \mathcal{H}_k^{\mathrm{c}}(x_k, x_{k-1}).\end{aligned}$$

For the second energy, put

$$\Phi_k = F + \frac{\mu_k}{2}\|\cdot\|^2, \qquad m_k = \min \Phi_k, \qquad z_k = \operatorname{argmin} \Phi_k, \qquad \Psi_k = \Phi_k - m_k. \qquad (97)$$

The regularized minimizer exists uniquely. Set

$$\begin{aligned}\mathcal{M}_k(x,p) &= d_k \Psi_k(x) + \frac{1}{2}\|x-p\|^2, \\ \mathcal{E}_k^{\mathrm{c}}(x,p) &= \mathcal{M}_k(x,p) + \tau_k\langle p-z_k, x-p+d_k[\nabla H(p)-\nabla H(z_k)]\rangle + \frac{\tau_k b_k}{2}\|p-z_k\|^2, \qquad (98)\\ e_k &= \mathcal{E}_k^{\mathrm{c}}(x_k, x_{k-1}).\end{aligned}$$

Theorem 4. *For the class and original initialization above, the following assertions hold.*

(i) *For every $k \geq 2$ and every independent phase $x \in \operatorname{dom} G$, $p \in \mathbb{R}^d$,*

$$\mathcal{H}_k^{\mathrm{c}}(x,p) \geq \frac{k}{4}(F(x)-F_*) + \frac{1}{8}\|x\|^2 + \frac{k}{4a_k}\|x-p\|^2 + \frac{91}{8192}\|p-z^*\|^2. \qquad (99)$$

With $K = 360000$, the actual trajectory satisfies

$$H_{k+1}^{\mathrm{c}} \leq \left(1 - \frac{1}{512\sqrt{k}}\right) H_k^{\mathrm{c}} + \frac{5Z^2}{512\sqrt{k}} \qquad (k \geq K). \qquad (100)$$

Consequently, with the finite constants

$$B = \max\{H_2^{\mathrm{c}}, \dots, H_K^{\mathrm{c}}, 5Z^2\}, \qquad X = \sqrt{8B}, \quad D = 2\sqrt{B}, \quad \Gamma = \sqrt{8B}, \qquad (101)$$

we have, for every $k \geq 2$,

$$\|x_k\| \leq X, \quad \|v_k\| \leq D/\sqrt{k}, \quad 0 \leq F(x_k) - F_* \leq 4B/k, \quad \|\nabla H(x_k) - \nabla H(z^*)\| \leq \Gamma/\sqrt{k}. \qquad (102)$$

(ii) *For every $k \geq 4$ and every such independent phase,*

$$\frac{3}{4}\mathcal{M}_k \leq \mathcal{E}_k^{\mathrm{c}} \leq 2\mathcal{M}_k, \qquad (103)$$

$$\begin{aligned}\mathcal{E}_k^{\mathrm{c}}(x,p) &\geq d_k \Psi_k(x) + \frac{3}{8}\|x-p\|^2 + \frac{\tau_k b_k}{4}\|p-z_k\|^2 \qquad (104)\\ &\geq \frac{1}{20k}\|x-z_k\|^2 + \frac{3}{8}\|x-p\|^2.\end{aligned}$$

If y, q are its literal kth update, then, with the old coefficients and center,

$$\mathcal{E}_k^c(q, x) \le (1 - \tau_k/4)\mathcal{E}_k^c(x, p). \tag{105}$$

Put

$$\begin{aligned} &P = X + Z, \qquad W = D + (\Gamma + Z)/4, \qquad a_0 = 1/256, \qquad N = 2^{20}, \\ &C_0 = \frac{3X^2}{16} + \frac{Z^2}{8} + \frac{PW}{128} + \frac{ZW}{64} + \frac{3P(\Gamma + Z)}{1024} + \frac{13PZ}{512}. \end{aligned} \tag{106}$$

The actual changing-index energies satisfy

$$e_{k+1} \le (1 - a_0/\sqrt{k})e_k + C_0/k^2 \qquad (k \ge 4). \tag{107}$$

(iii) *Let $Q = \max\{N^{3/2}e_N, 2C_0/a_0\}$. For every $k \ge N$,*

$$\begin{aligned} &e_k \le Qk^{-3/2}, \qquad \|x_k - z_k\|^2 \le 20Q/\sqrt{k}, \qquad \|v_k\|^2 \le 8Q/(3k^{3/2}), \\ &0 \le F(x_k) - F_* \le 10Q/k^{3/2} + Z^2/(2k). \end{aligned} \tag{108}$$

Moreover,

$$x_k \longrightarrow z^*, \qquad y_k \longrightarrow z^*, \qquad \sum_{k\ge1} \|x_k - x_{k-1}\|^2 < \infty. \tag{109}$$

The first energy controls the original phase before any moving-center estimate is used. The constants K, N do not prescribe a restart or an additional algorithmic operation. No bounded normal selection, strong convexity or coercivity of F, or boundedness of S is assumed. The position rate in (108) is relative to z_k; no universal norm rate to z^* or finite-length assertion is made. The method, affine rebalancing, cocoercivity and Tikhonov-path tools are classical. The source comparison uses the author-uploaded final text of [7]; its final typeset PDF was not available for visual comparison. The related source scopes and version limits are recorded in the accompanying verification guide. We make no firstness or maximal-range claim.

10 Proof of Theorem 4

10.1 Domains and a common proximal calculation

The smooth sum rule gives $-\nabla H(z^*) \in \partial G(z^*)$. Smooth convexity and the subgradient inequality therefore imply

$$\frac{1}{2}\|\nabla H(x) - \nabla H(z^*)\|^2 \le F(x) - F_* \qquad (x \in \operatorname{dom} G). \tag{110}$$

The proximal objective is strongly convex and coercive for every finite input and positive d_k, using an affine lower bound for G. Thus all original steps exist uniquely and $x_k \in \operatorname{dom} G$ for $k \ge 2$. This establishes finite-domain entry without any energy estimate.

We first give a calculation used for both energies. Freeze k, abbreviate its scalar coefficients by t, μ, a, d, b, τ, and consider convex functions h, g both minimized at a point z, where h is 1-smooth. Fix a vector c and perform

$$y = x + a(x - p) - d[\nabla h(x) - \nabla h(p)], \qquad q = \operatorname{prox}_{dg}(y - d[\nabla h(y) + \mu(x - c)]).$$

Write $v = x - p$, $u = y - x$, $w = q - x$, $r = y - q$, and

$$J(x) = h(x) - h(z) + g(x) - g(z), \quad M(x, p) = d[J(x) + \mu\|x - c\|^2/2] + \|v\|^2/2,$$

$$T(x, p) = \langle p - z, v + d\nabla h(p)\rangle + b\|p - z\|^2/2, \qquad E = M + \tau T.$$

Here $x \in \operatorname{dom} g$ and p is arbitrary; the predictor need not be feasible. Proximal optimality selects

$$\xi = r/d - \nabla h(y) - \mu(x - c) \in \partial g(q).$$

Define the nonnegative remainders

$$\begin{aligned} D_h(s, y) &= h(s) - h(y) - \langle \nabla h(y), s - y\rangle, \\ R_h(q, y) &= \|q - y\|^2/2 - D_h(q, y), \\ B_g(s; q) &= g(s) - g(q) - \langle \xi, s - q\rangle \qquad (s = x, z). \end{aligned} \tag{111}$$

In a frozen difference, direct expansion yields

$$\begin{aligned} \Delta M = {} & \tfrac{1}{2}(\|u\|^2 - \|v\|^2) - \tfrac{1-d}{2}\|r\|^2 + \tfrac{d\mu}{2}\|w\|^2 \\ & - dD_h(x, y) - dR_h(q, y) - dB_g(x; q), \end{aligned} \tag{112}$$

$$\Delta T = (1 - b/2)\|v\|^2 + d\langle v, \nabla h(p)\rangle - \langle x - z, r\rangle, \tag{113}$$

$$\begin{aligned} -\langle x - z, r\rangle = {} & - dJ(x) - d\mu\langle x - z, x - c\rangle \\ & + dD_h(x, y) - dD_h(z, y) + dB_g(x; q) - dB_g(z; q). \end{aligned} \tag{114}$$

For example, the last identity converts the nonsmooth subgradient inequality at the actual output q into a current-point objective difference. Its positive $B_g(x; q)$ is canceled by the larger negative term in (112). No norm estimate for ξ is used.

The average Hessian of h between p, x lies between 0 and I, whence

$$\|u\| \le a\|v\|, \qquad \|w\|^2 \le 2a^2\|v\|^2 + 2\|r\|^2, \qquad d\langle v, \nabla h(p)\rangle \le \tfrac{d}{2}J(x) + 2d\|v\|^2.$$

The last bound follows from $\|\nabla h(p)\| \le \sqrt{2J(x)} + \|v\|$. Put

$$C_v = (1 - a^2)/2 - d\mu a^2 - \tau(1 - b/2 + 2d), \qquad C_r = (1 - d)/2 - d\mu.$$

For $0 < t \le 1$, exact simplification gives

$$\begin{aligned} C_v - b/2 - \tau/4 &= \frac{t(1480 + 1588t - 234t^2 - 81t^3)}{256(3t + 2)^3} > 0, \\ C_r - 1/4 &= \frac{1 + 3t - 2t^2}{4(3t + 2)} > 0. \end{aligned} \tag{115}$$

Indeed the first parenthesis is at least 1165, and the second numerator is $1+t+2t(1-t)$. Combining the three identities, and using

$$2\langle x-z, x-c\rangle = \|x-c\|^2 + \|x-z\|^2 - \|z-c\|^2,$$

proves the common estimate

$$\Delta E \le -\frac{\tau}{2}M - \frac{b}{2}\|v\|^2 - \frac{1}{4}\|r\|^2 - \frac{\tau d\mu}{4}\|x-c\|^2 - \frac{\tau d\mu}{2}\|x-z\|^2 + \frac{\tau d\mu}{2}\|z-c\|^2. \quad (116)$$

All discarded remainder terms have coefficients $-d(1-\tau), -d, -d(1-\tau), -\tau d, -\tau d$, respectively, on $D_h(x,y), R_h(q,y), B_g(x;q), D_h(z,y), B_g(z;q)$. They are nonpositive throughout the schedule.

10.2 A fixed composite minimizer controls the original phase

Apply the calculation with

$$z = z^*, \quad c = 0, \quad h = H - \langle \nabla H(z^*), \cdot\rangle, \quad g = G + \langle \nabla H(z^*), \cdot\rangle.$$

The balanced components are minimized at z^*, and the proximal update is unchanged. Moreover $(k/a_k)E = \mathcal{H}^{\mathrm{c}}_k$ exactly. Monotonicity of ∇h keeps its pairing nonnegative, while

$$\frac{\sqrt{k}}{64a_k}\,|\langle p-z^*, x-p\rangle| \le \frac{k}{4a_k}\|x-p\|^2 + \frac{1}{4096a_k}\|p-z^*\|^2.$$

Since $a_k \ge 2/5$, this proves (99).

For $k \ge 4$ let $P_k = (k/a_k)M$ and put $U = \|p-z^*\|$, $e = \sqrt{k}\|x-p\|$, $f = \sqrt{k(F(x)-F_*)}$. The bounds $a_k \ge 4/7$ and (110) give

$$\mathcal{H}^{\mathrm{c}}_k - P_k \le Ue/32 + \sqrt{2}Uf/256 + 3U^2/256 \le U^2/32 + e^2/64 + f^2/512.$$

Since $U^2 \le 4\|x\|^2 + e^2 + 2Z^2$, the last expression is at most $P_k + Z^2/16$. Thus

$$\mathcal{H}^{\mathrm{c}}_k \le 2P_k + Z^2/16. \quad (117)$$

Multiplying (116) by k/a_k and using this upper comparison gives the frozen bound

$$\mathcal{H}^{\mathrm{c}}_k(q,x) \le (1-\tau_k/4)\mathcal{H}^{\mathrm{c}}_k(x,p) - \frac{kb_k}{2a_k}\|v\|^2 - \frac{k}{4a_k}\|r\|^2 + \frac{9\tau_k}{64}Z^2. \quad (118)$$

We next change its coefficients at the new phase. Let $c_k = (k+1)/k$ and

$$\eta_{0,k} = \frac{(k+1)/\sqrt{k} - \sqrt{k+1}}{64}, \qquad \eta_{1,k} = \eta_{0,k} + \frac{3}{128k}.$$

Exact subtraction gives

$$\begin{aligned}
&\mathcal{H}^{\mathrm{c}}_{k+1}(q,x) - c_k\mathcal{H}^{\mathrm{c}}_k(q,x) \\
&= -\frac{\|q\|^2}{8k} + \frac{k+1}{2}(a_{k+1}^{-1} - a_k^{-1})\|w\|^2 - \eta_{1,k}\langle x-z^*, w\rangle \\
&\quad - \frac{\eta_{0,k}}{4}\langle x-z^*, \nabla H(x) - \nabla H(z^*)\rangle - \frac{3}{256k}\|x-z^*\|^2.
\end{aligned} \quad (119)$$

The kinetic and gradient terms are nonpositive. For $k \geq 4$, $0 \leq \eta_{0,k} \leq 1/(64\sqrt{k})$ and $\eta_{1,k} \leq 7/(256\sqrt{k})$. Completing the square bounds the remaining mixed and last terms by

$$\frac{64k}{3}\eta_{1,k}^2\|w\|^2 \leq \frac{49}{3072}\|w\|^2 \leq \frac{1}{24}(a_k^2\|v\|^2 + \|r\|^2).$$

These costs are absorbed by the losses in (118), since $kb_k/(2a_k) = 3\sqrt{k}/4$ and $k/(4a_k) \geq k/4$. Hence

$$H_{k+1}^c \leq c_k(1 - \tau_k/4)H_k^c + c_k 9\tau_k Z^2/64.$$

For $k \geq 360000$ its coefficients satisfy (100). Every $H_2^c, \ldots, H_K^c$ is finite by the already established domain entry. Induction in the last recurrence and coercivity prove (102), including the balanced-gradient bound from (110). Boundedness has thus been obtained before moving the comparator.

10.3 The composite reference path and its gradient

The function Φ_k is strongly convex and bounded below by $F_* + \mu_k\|\cdot\|^2/2$, so z_k exists uniquely. Comparison with z^* gives

$$\|z_k\| \leq Z, \quad 0 \leq F(z_k) - F_* \leq Z^2/(2k), \quad \|\nabla H(z_k) - \nabla H(z^*)\| \leq Z/\sqrt{k}. \tag{120}$$

The last estimate uses (110). For adjacent centers write $z = z_k$, $z' = z_{k+1}$, $\mu = 1/k$, $\mu' = 1/(k+1)$, $e = z' - z$ and $A = \nabla H(z') - \nabla H(z)$. The actual optimality inclusions are

$$-\nabla H(z) - \mu z \in \partial G(z), \qquad -\nabla H(z') - \mu' z' \in \partial G(z').$$

Subtracting them, monotonicity of ∂G and cocoercivity yield

$$\|A\|^2 + \mu\|e\|^2 \leq (\mu - \mu')\langle z', e\rangle \leq (\mu - \mu')Z\|e\|. \tag{121}$$

Completing the square, and treating $e = 0$ separately for the displacement, gives

$$\|z_{k+1} - z_k\| \leq \frac{Z}{k+1}, \qquad \|\nabla H(z_{k+1}) - \nabla H(z_k)\| \leq \frac{Z}{2\sqrt{k}(k+1)}. \tag{122}$$

No bound on either selected nonsmooth subgradient is used. Every cluster point of the bounded path lies in S by lower semicontinuity and (120), and its norm is at most Z. It equals z^*; finite-dimensional compactness therefore gives $z_k \to z^*$. This argument does not require continuity of G on its domain boundary.

10.4 The centered frozen energy

Fix $k \geq 4$ and apply the common calculation with

$$z = z_k, \quad c = z_k, \quad h = H - \langle \nabla H(z_k), \cdot\rangle, \quad g = G + \langle \nabla H(z_k) + \mu_k z_k, \cdot\rangle.$$

The two components are minimized at z_k. Their sum is $H+G+\langle \mu_k z_k, \cdot \rangle$, and consequently

$$J(x) + \mu_k \|x - z_k\|^2/2 = \Psi_k(x).$$

The shift of the proximal affine term cancels the shift from $\mu_k(x - z_k)$, so the actual map is still (95). Here $M = \mathcal{M}_k$, $E = \mathcal{E}_k^{\mathrm{c}}$, and the forcing in (116) is zero.

Since $\tau_k/b_k = 1/(96a_k) \le 5/192$, square completion proves (104) and its lower comparison with $3\mathcal{M}_k/4$. For the upper comparison, write $f = \sqrt{\Psi_k(x)}$, $e = \|x - p\|$, $U = \|p - z_k\|$. Strong convexity and the balanced smooth gap give

$$U \le \sqrt{2} f/t_k + e, \qquad \|\nabla h(p)\| \le \sqrt{2} f + e.$$

Expansion of the three added terms in the energy yields

$$E - M \le \frac{d}{8} f^2 + Cfe + De^2, \qquad C = \frac{\sqrt{2}}{64}[1 + d(1 + t) + b], \quad D = \frac{t(1 + d)}{64} + \frac{3at^2}{256}.$$

For $0 < t \le 1$, $C \le 9\sqrt{2}/320$ and $D \le 1/32$. Thus $Cfe \le df^2/4 + 81e^2/5120$, and

$$E \le \frac{11d}{8} f^2 + \frac{2801}{5120} e^2 \le 2M.$$

This proves (103) on the whole phase domain. The common descent now gives $\Delta E \le -\tau_k \mathcal{M}_k/2 \le -\tau_k E/4$, which is (105). Its comparator and coefficients are still frozen; no changing-index conclusion has yet been taken.

10.5 All changes of the center and coefficients

At the new phase write $p' = x_k$, $x' = x_{k+1}$, $v' = x' - p'$, and

$$r_j = p' - z_j, \qquad w_j = v' + d_j[\nabla H(p') - \nabla H(z_j)], \qquad j = k, k + 1.$$

Equations (102) and (120) give

$$\|r_k\|, \|r_{k+1}\| \le P, \qquad \|w_k\| \le W/\sqrt{k}, \tag{123}$$

with exactly P, W from (106). The scalar schedules satisfy

$$\begin{aligned} 0 < d_{k+1} - d_k \le 3/(16k^{3/2}), \qquad & |\tau_{k+1} - \tau_k| \le 1/(128k^{3/2}), \\ c_k := \tau_k b_k/2 = \frac{3}{256(k + 3\sqrt{k}/2)}, \qquad & c_{k+1} \le c_k. \end{aligned} \tag{124}$$

These follow by differentiation of the displayed scalar functions and integration over $[k, k+1]$.

Put $\Delta\mu_k = 1/[k(k + 1)]$. Evaluating Φ_k at z_{k+1} gives $m_k - m_{k+1} \le \Delta\mu_k Z^2/2$. Hence

$$\Psi_{k+1}(x') - \Psi_k(x') = -\Delta\mu_k \|x'\|^2/2 + m_k - m_{k+1} \le \Delta\mu_k Z^2/2.$$

Also $\Psi_{k+1}(x') \le X^2/(k+1)$ by the first-stage bounds. Splitting the product using the increasing d_k therefore gives

$$d_{k+1}\Psi_{k+1}(x') - d_k\Psi_k(x') \le (3X^2/16 + Z^2/8)/k^2. \tag{125}$$

The kinetic term is unchanged. The complete mixed-term difference is

$$\begin{aligned}&\tau_{k+1}\langle r_{k+1}, w_{k+1}\rangle - \tau_k\langle r_k, w_k\rangle\\ &= (\tau_{k+1}-\tau_k)\langle r_k, w_k\rangle + \tau_{k+1}\langle r_{k+1}-r_k, w_k\rangle + \tau_{k+1}\langle r_{k+1}, w_{k+1}-w_k\rangle.\end{aligned} \tag{126}$$

Its first two upper costs are $PW/(128k^2)$ and $ZW/(64k^2)$. For the third use the exact vector difference

$$w_{k+1} - w_k = (d_{k+1}-d_k)[\nabla H(p') - \nabla H(z_k)] - d_{k+1}[\nabla H(z_{k+1}) - \nabla H(z_k)].$$

The two-inclusion estimate (122) then bounds its contribution by

$$\big[3P(\Gamma + Z)/1024 + PZ/512\big]/k^2.$$

In particular no unestimated nonsmooth normal enters this cost. Finally the decreasing c_k and the difference-of-squares identity give

$$c_{k+1}\|r_{k+1}\|^2 - c_k\|r_k\|^2 \le c_{k+1}(\|r_{k+1}\| + \|r_k\|)\|z_{k+1} - z_k\| \le \frac{3PZ}{128k^2}.$$

Adding these costs proves $e_{k+1} \le \mathcal{E}_k^{\mathrm{c}}(x_{k+1}, x_k) + C_0/k^2$. Together with (105), this is the actual recurrence (107).

10.6 Decay and the original trajectory limit

The definition of Q gives $e_N \le QN^{-3/2}$ and $C_0 \le a_0 Q/2$. For $k \ge N$, $\sqrt{k} \ge 1024 \ge 3/a_0$, so

$$1 - \frac{a_0}{2\sqrt{k}} \le 1 - \frac{3}{2k} \le \left(\frac{k}{k+1}\right)^{3/2}.$$

Induction in (107) yields $e_k \le Qk^{-3/2}$, including $Q = 0$ without division by Q. The lower energy bounds give the position and velocity estimates in (108). In particular $ke_k \to 0$; convergence is not inferred from $e_k \to 0$ alone. The exact identity

$$F(x_k) - F_* = \Psi_k(x_k) + m_k - F_* - \mu_k\|x_k\|^2/2$$

and $m_k \le F_* + \mu_k Z^2/2$ give the displayed objective estimate. Now $x_k - z_k \to 0$ and $z_k \to z^*$ imply $x_k \to z^*$. The bound $\|y_k - x_k\| \le a_k\|v_k\|$ gives $y_k \to z^*$. The squared increments are summable on the tail by (108); the finite original prefix, including $x_1 - x_0$, adds finitely many finite terms.

If $B = 0$, coercivity at $k = 2$ gives $x_2 = 0$ and $x_2 - x_1 = 0$, and all later x_k are zero. The original x_0 need not vanish: the first proximal step may erase it. If $Z = 0$ but $B > 0$, all the nonzero B-dependent costs in C_0 remain. No finite value of $G(y_k)$, bounded normal, or replacement of the original initialization is used. This completes the proof.

References

[1] S. Qu, M. Staudigl and J. Peypouquet, *Inertial forward-backward algorithm with exterior penalization and Tikhonov regularization*, 2026. arXiv:2607.09495v1.

[2] H. Attouch, M.-O. Czarnecki and J. Peypouquet, *Prox-penalization and splitting methods for constrained variational problems*, SIAM Journal on Optimization **21** (2011), 149–173. doi:10.1137/100789464.

[3] A. Balhag, Z. Mazgouri and M. Théra, *Convergence of inertial prox-penalization and inertial forward-backward algorithms for solving bilevel monotone equilibrium problems*, HAL:hal-04137497v1, 2023. hal-04137497v1. The journal article appeared under the title ending "monotone bilevel equilibrium problems", Optimization **74** (2025), 2885–2929; doi:10.1080/02331934.2024.2341934. The comparison here is with the accessible author versions.

[4] S. C. László, *On the convergence of an inertial proximal algorithm with a Tikhonov regularization term*, Communications in Nonlinear Science and Numerical Simulation **149** (2025), 108924. doi:10.1016/j.cnsns.2025.108924. The cited theorem and its proof are those of the final journal article; the earlier preprint is arXiv:2302.02115v4.

[5] H. Attouch, A. Cabot, Z. Chbani and H. Riahi, *Inertial forward-backward algorithms with perturbations: application to Tikhonov regularization*, Journal of Optimization Theory and Applications **179** (2018), 1–36. doi:10.1007/s10957-018-1369-3. The lemma numbering cited here is from the author manuscript dated 27 August 2017, under the title *Accelerated forward-backward algorithms with perturbations*.

[6] H.-K. Xu, *Iterative algorithms for nonlinear operators*, Journal of the London Mathematical Society (2) **66** (2002), 240–256. doi:10.1112/S0024610702003332.

[7] H. Attouch, A. Balhag, Z. Chbani and H. Riahi, *Accelerated gradient methods combining Tikhonov regularization with geometric damping driven by the Hessian*, Applied Mathematics & Optimization **88** (2023), 29. doi:10.1007/s00245-023-09997-x. The IGATH-2 locator is Section 4.2, p. 45; IPGATH-1 is in Section 5.2, pp. 48–49. Both comparisons use the complete author-posted final text.

[8] S. Adly, V. T. Ho and H. N. Nguyen, *Accelerated inertial gradient algorithms with vanishing Tikhonov regularization*, 2026. arXiv:2601.23035v1.

[9] X. Sun, G. Tian and H. Zhang, *Convergence analysis of a Tikhonov regularized inertial dynamical system and algorithm for convex optimization problems*, arXiv:2506.15968v3, 2026. The related journal article is linked at doi:10.1007/s11075-026-02386-9; the comparison here is with the indicated author version.

[10] H. Attouch, Z. Chbani, J. Fadili and H. Riahi, *First-order optimization algorithms via inertial systems with Hessian driven damping*, Mathematical Programming **193** (2022), 113–155. doi:10.1007/s10107-020-01591-1. The inspected algorithm and theorem are in arXiv:1907.10536v2.

[11] H. Attouch and J. Fadili, *From the Ravine method to the Nesterov method and vice versa: a dynamical system perspective*, SIAM Journal on Optimization **32** (2022), 2074–2101. doi:10.1137/22M1474357. The exact interchange was inspected in arXiv:2201.11643v2 and the later author-hosted revision.

[12] H. Attouch, J. Fadili and V. Kungurtsev, *The stochastic Ravine accelerated gradient method with general extrapolation coefficients*, arXiv:2403.04860v2, 2024. The journal article is Journal of Optimization Theory and Applications **209** (2026), 5, doi:10.1007/s10957-026-02940-4; the theorem comparison uses the indicated author version.